White Box Testing Practice Questions and Solution

CLO #	Course Learning Outcomes (CLOs)	PLO Mapping	Bloom's Taxonomy
CLO 2	**Write** and **Use** test cases to evaluate functional and non-functional specification requirements for various test types /levels / techniques and types of information systems and technologies.	PLO_4 (Design/ Development of Solutions)	C3 (Applying)

Q1.

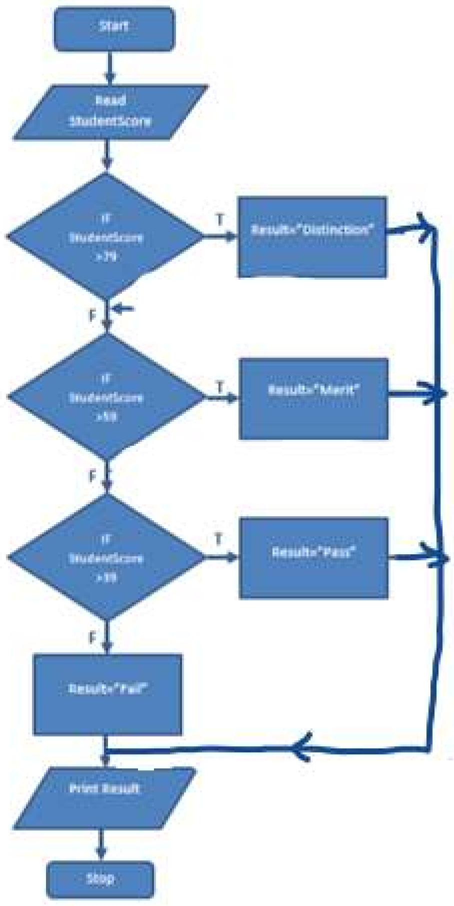
For the following program answer following.

a. **Construct** a control flow chart.
b. **Construct** a control flow graph.
c. **Construct** a hybrid flow graph.
d. **Discover** how many test cases would be needed for 100 per cent statement coverage? Also **Compute** whether 100% statement coverage is achieved if: 'StudentScore = 50' and 'StudentScore = 30'. If not, **identify** which lines of pseudo code will not be exercised?

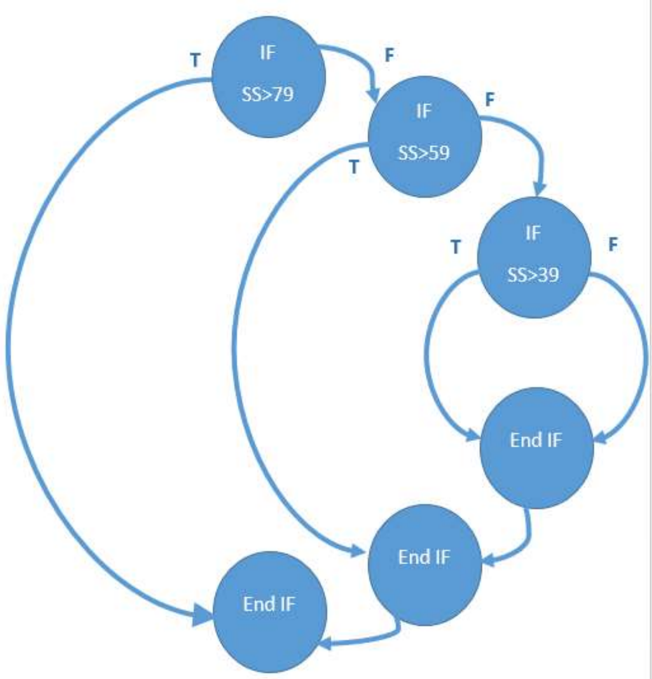
```
1   Program Grading
2
3   StudentScore: Integer
4   Result: String
5
6   Begin
7
8   Read StudentScore
9
10  If StudentScore > 79
11  Then Result = "Distinction"
12  Else
13      If StudentScore > 59
14      Then Result = "Merit"
15      Else
16          If StudentScore > 39
17          Then Result = "Pass"
18          Else Result = "Fail"
19          Endif
20      Endif
21  Endif
22  Print ("Your result is", Result)
23  End
```

Answer:

(a) Control flow chart:

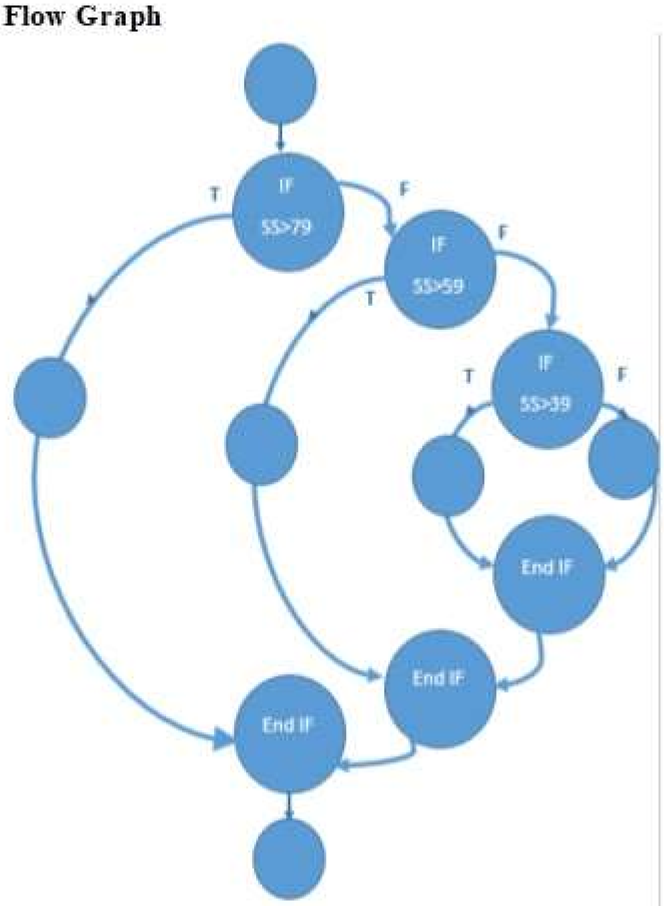


(b) Control flow graph:



(c) Hybrid flow graph:

Answer 1 (c) Hybrid Flow Graph



(d) 4 test cases would be required for100 per cent statement coverage

- No100% statement coverage is not achieved if ‘StudentScore = 50’
- Line number 11, 14 and 18 would not be exercised.

4 test cases would be required for100 per cent statement coverage

- No100% statement coverage is not achieved if ‘StudentScore = 30’
- Line number 11, 14 and 17 would not be exercised.

Q2.

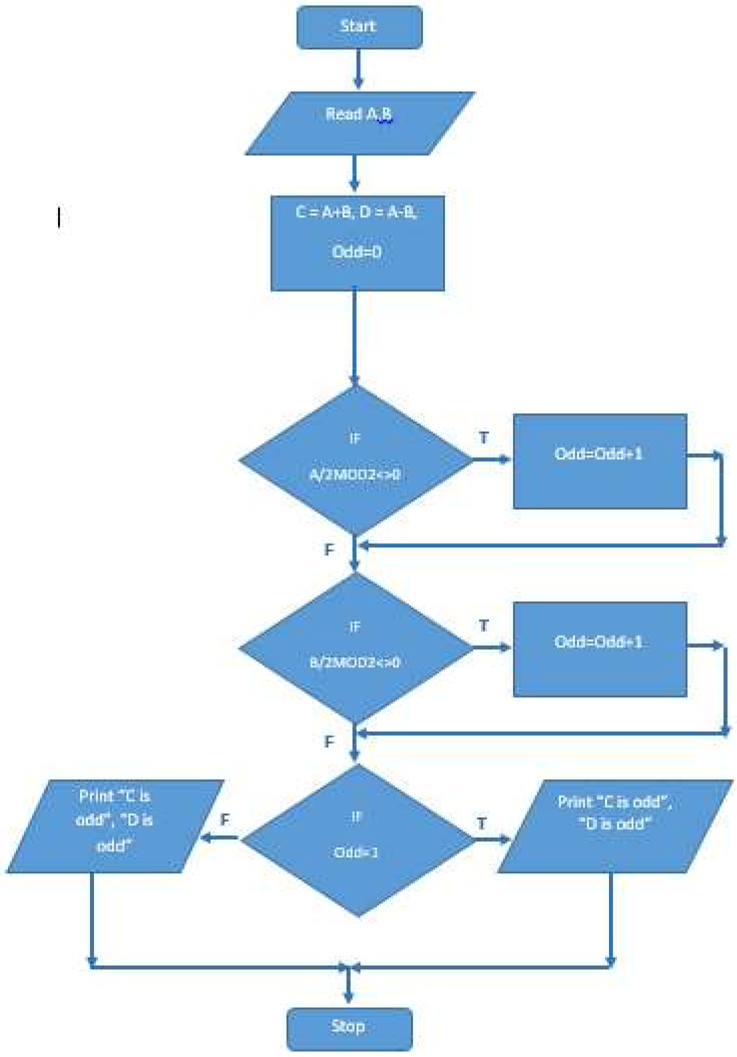
For the following program answer following.

a. **Construct** a control flow chart.
b. **Construct** a control flow graph.
c. **Construct** a hybrid flow graph.
d. **Write** test data for maximum statement coverage.
e. **Write** test data for maximum decision coverage.

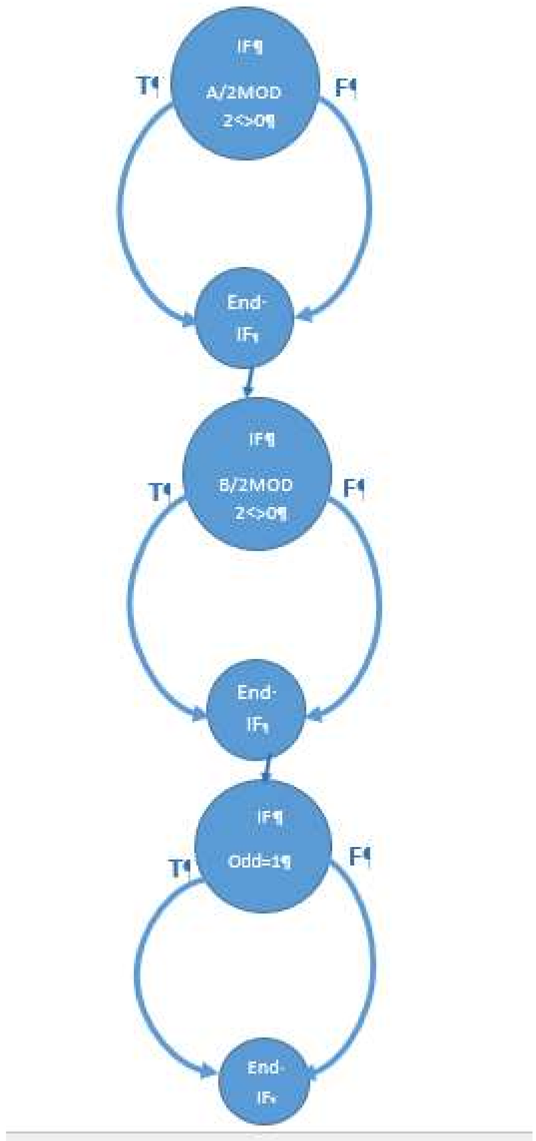
```
Program OddandEven

A, B: Real;
Odd: Integer;

Begin
    Read A
    Read B
    C = A + B
    D = A – B
    Odd = 0

    If A/2 DIV 2 <> 0 (DIV gives the remainder after division)
    Then Odd = Odd + 1
    Endif

    If B/2 DIV 2 <> 0
    Then Odd = Odd + 1
    Endif

    If Odd = 1
    Then
        Print ("C is odd")
        Print ("D is odd")
    Else
        Print ("C is even")
        Print ("D is even")
    Endif
End
```

Answer:

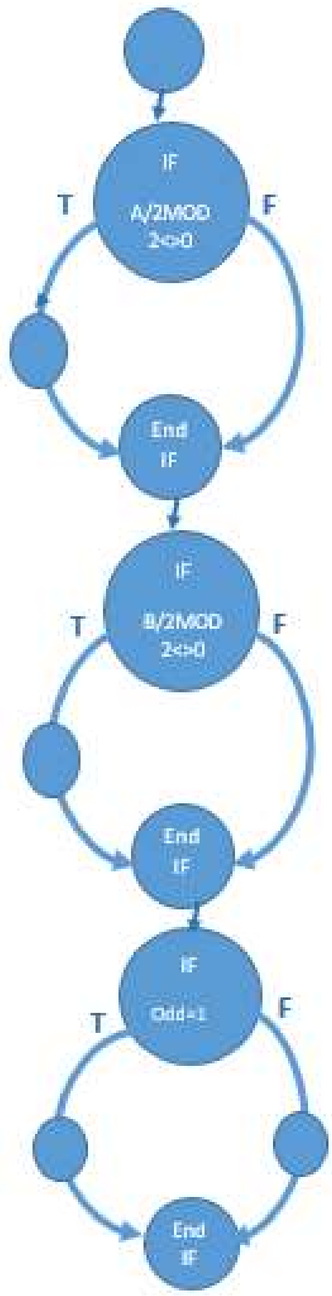
a. **Construct** a control flow chart.



B **Construct** a control flow graph.



C **Construct** a hybrid flow graph.



d. **Write** test data for maximum statement coverage.

- Test Data for maximum Statement coverage is

A	B
4	4
5	5

e. **Write** test data for maximum decision coverage.

A	B	Odd	Path exercised	Comments
4	4	0	FFF	Possible
4	4	0	FFT	Not possible
4	5	1	FTF	Not possible
4	5	1	FTT	Possible
5	4	1	TFF	Not possible
5	4	1	TFT	Possible
5	5	2	TTF	Not possible
5	5	2	TTT	Possible

Test Data for maximum Decision Coverage is:

A	B
4	4
4	5
5	4
5	5

Q3.

For the following program answer following.

a. **Construct** a control flow chart.

b. **Construct** a control flow graph.

c. **Construct** a hybrid flow graph.

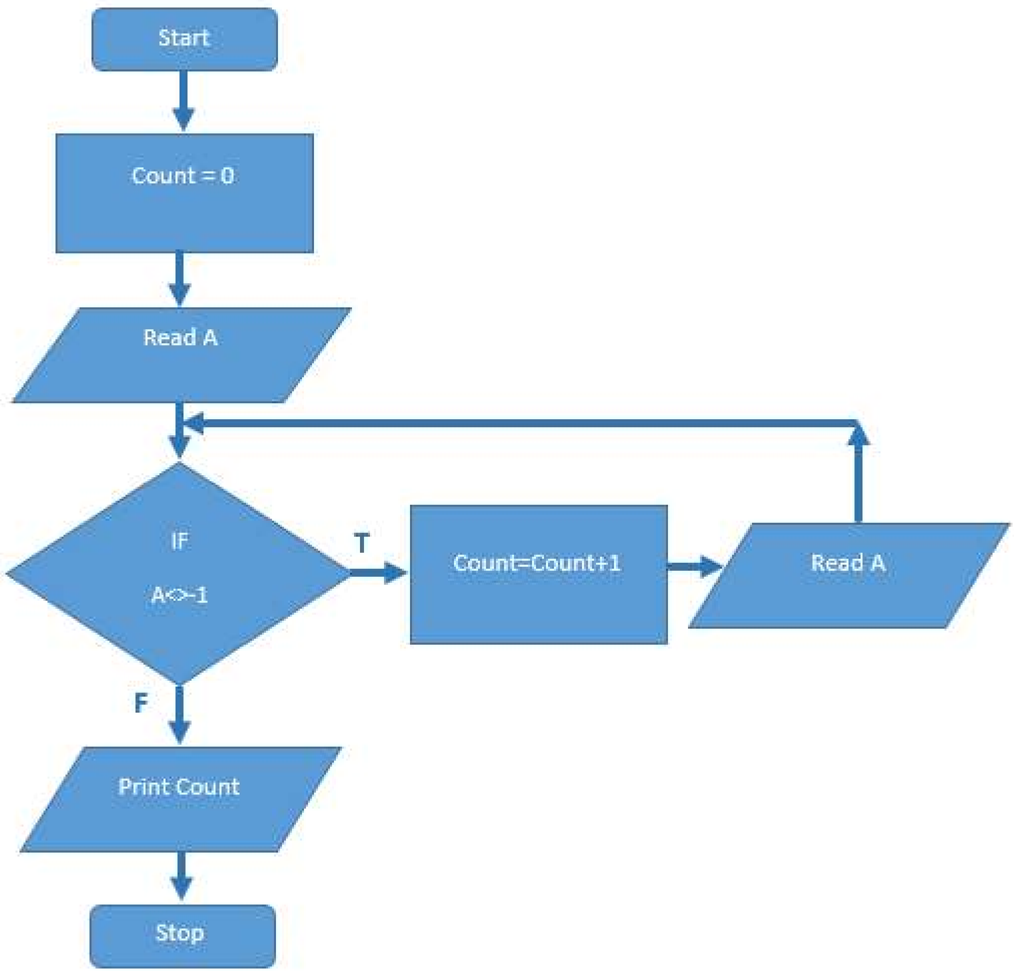
d. **Discover** how many test cases are needed to achieve 100 per cent decision coverage?

e. **Compute** what level of decision coverage would be achieved by the single input A = –1?

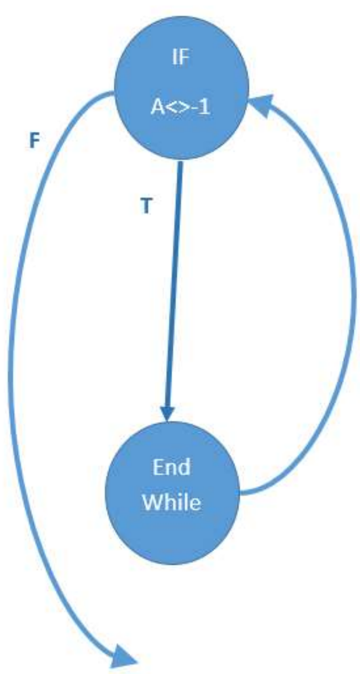
```
Program Counting numbers
A: Integer
Count: Integer
Begin
Count = 0
Read A
While A <> –1
Do
    Count = Count + 1
    Read A
EndDo
Print ("There are", Count, "numbers in the list")
End
```

Answer:

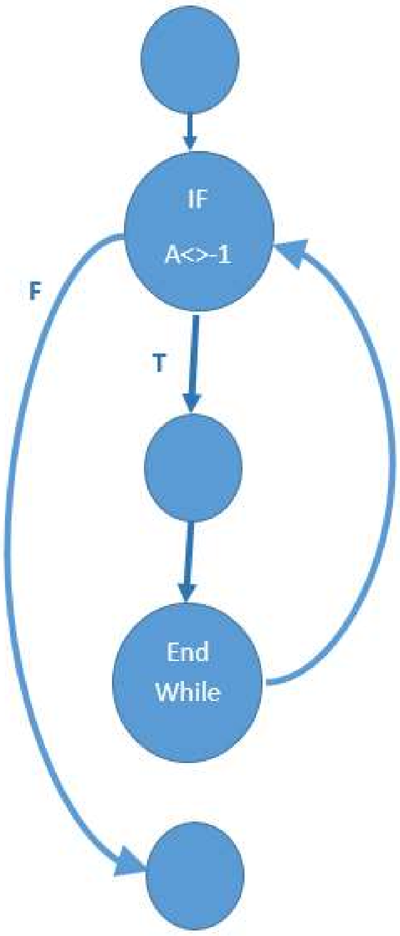
A **Construct** a control flow chart.



b. **Construct** a control flow graph.



c. **Construct** a hybrid flow graph.



D **Discover** how many test cases are needed to achieve 100 per cent decision coverage?

2 test cases would be required for100 per cent decision coverage

e. **Compute** what level of decision coverage would be achieved by the single input A = –1?

Since,

Decision Coverage = Total No. of decision outcomes exercised/Total No. of decision outcomes *100

Therefore,

For A=-1, only the false path would be executed.

⇨ Total No. of decision outcomes exercised = 1

Substituting values,

⇨ Decision Coverage =1/2*100

⇨ Decision Coverage =50%